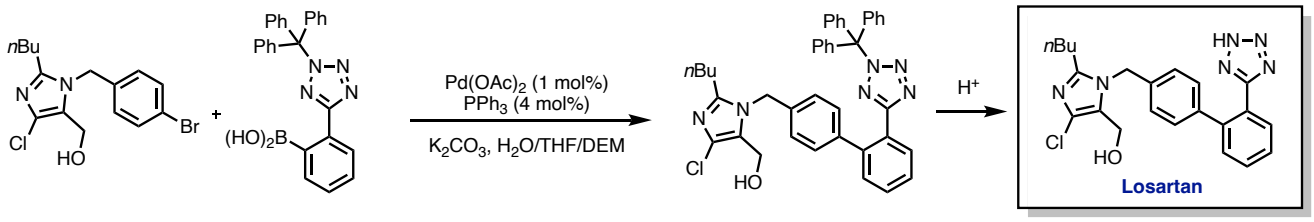
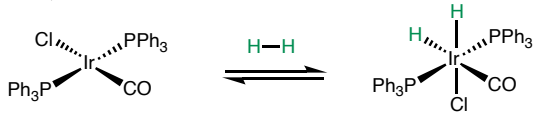
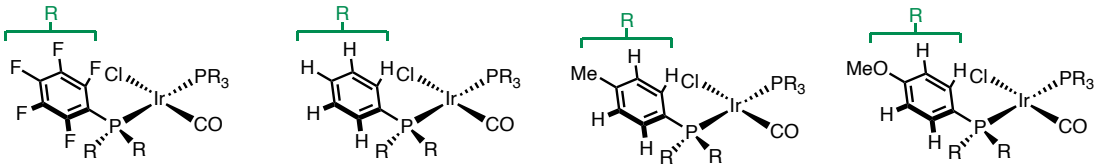
1. Indicate which coupling partner reacts as nucleophile and which one as electrophile in the following Suzuki coupling (*J. Org. Chem.* **1994**, *59*, 6391−6394) [★] [DEM, diethoxymethane]



2. Suggest a possible spectroscopic technique to follow the oxidative addition of dihydrogen on Vaska's complex (*J. Am. Chem. Soc.* **1962**, *84*, 679–680). Explain why the observed kinetic isotope effect when D2 is used is only $\frac{k_H}{k_D} = 1.22$ (*J. Am. Chem. Soc.* **1966**, *88*, 3511–3514). [★]

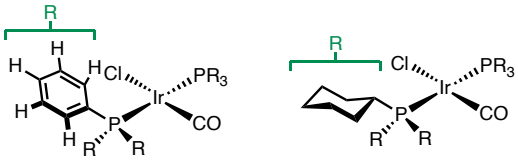


3. Rationalize the following bimolecular kinetic constants (k_2) for the for the oxidative addition of H_2 with the following Vaska's-type complexes (*Ann. N. Y. Acad. Sci.* **1971**, *172*, 546-562). [★]



k_2 = n.d. | k_2 = 1.2 s^{-1} M^{-1} | k_2 = 1.7 s^{-1} M^{-1} | k_2 = 2.1 s^{-1} M^{-1}

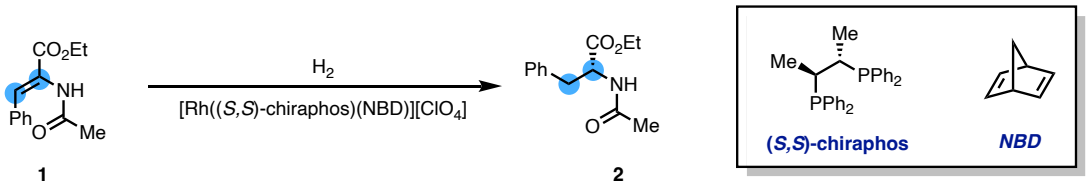
4. Explain the following seemingly contradictory data trends, where k_2 is the bimolecular kinetic constant for the for the oxidative addition of H_2 and υ_{CO} is the stretching frequency of the carbon monoxide ligand (*Ann. N. Y. Acad. Sci.* **1971**, *172*, 546-562). [★★]

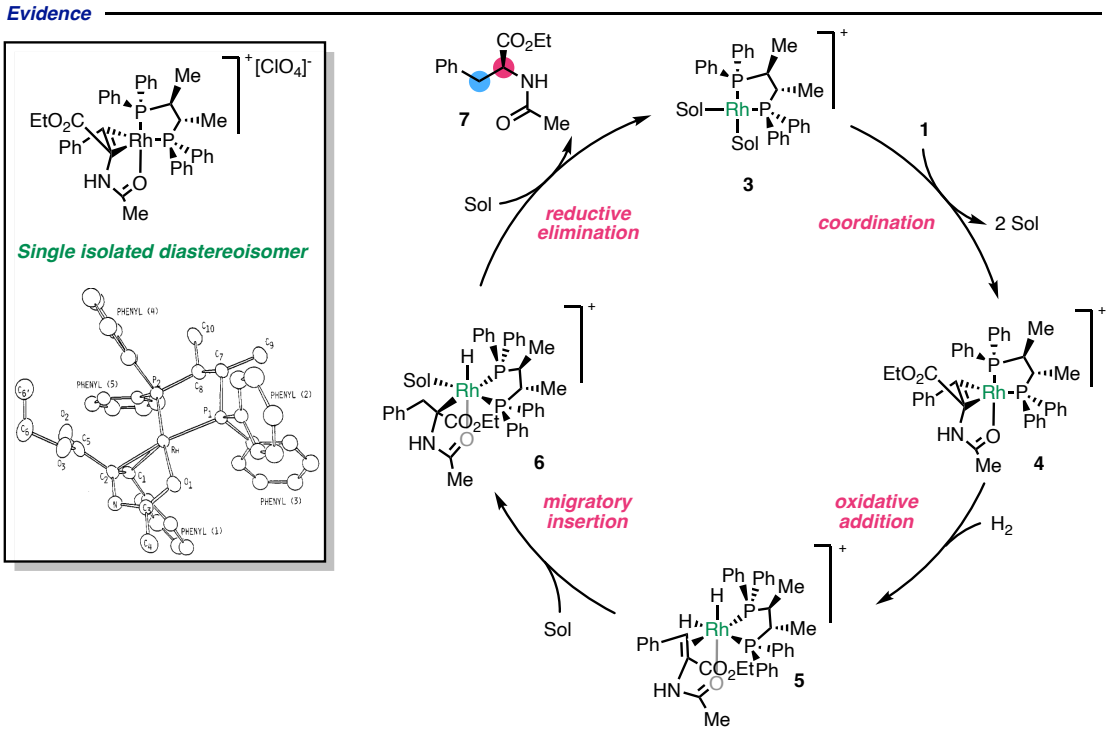


k_2 = 1.2 s^{-1} M^{-1} | k_2 = 6.6 10^{-3} s^{-1} M^{-1}

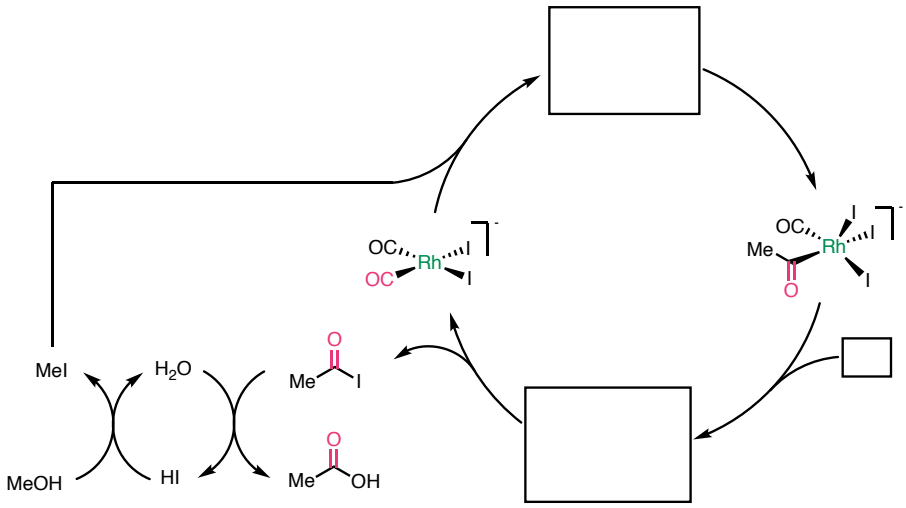
υ_{CO} = 1954 cm^{-1} | υ_{CO} = 1934 cm^{-1}

5. Halpern and co-workers studied the Rh-catalyzed asymmetric hydrogenation of prochiral acetoamidocynnamates (*J. Am. Chem. Soc.* **1980**, *102*, 5952-5954) and found the following: (a) The authors observed in solution and isolated a single diastereoisomer (**4**) from alkene coordination, but if the commonly accepted mechanism is followed (migratory insertion, *syn*-reductive elimination) the opposite enantiomer (**7**) of the observed product (**2**) would be generated; (b) The rate law reaction is $-\frac{d[H_2]}{dt} = k_{ox.add.}[\mathbf{4}][H_2]$, which tells that the H_2 oxidative addition is the rate determining step. Rationalize the stereochemical outcome of the reaction. (This system operates analogously to the DIPAMP-based Knowles hydrogenation, see *Angew. Chem. Int. Ed.* **2002**, *41*, 1998-2007) [★★]



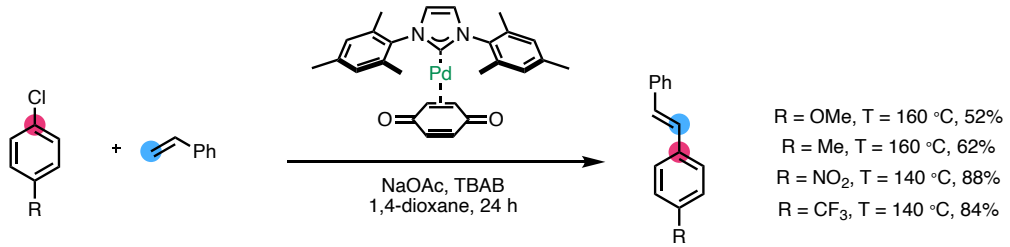


6. Complete the proposed catalytic cycle of the Monsanto process (acetic acid synthesis, for an overview: Hartwig, J. F. Organotransition Metal Chemistry, from Bonding to Catalysis; University Science Books: New York, 2010). Explain why for rhodium (Monsanto process) the oxidative addition is rate-determining, while for the BP-Cativa process (based on iridium) is the migratory insertion step. [★★]

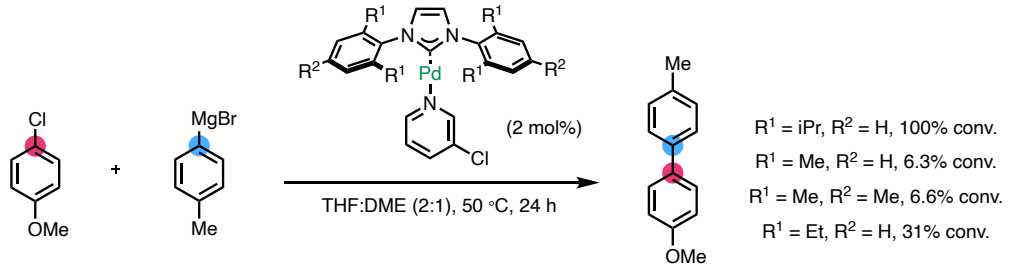


7. In 2002, Umicore patented (World Patent 2004, 014, 550, 2003) the following Pd-carbene pre-catalyst for the Mizokori-Heck coupling of aryl chlorides. Explain the ligand role and hypothesize the reason behind the yields trend (higher for

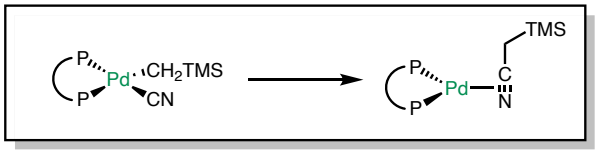
aryl chloride with electron-withdrawing substituents and required temperature). [TBAB, tetrabutylammonium bromide] [★]

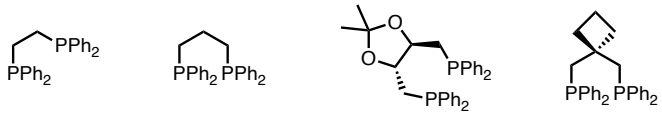


8. Explain the following trends in the Kumada coupling using PEPPSI-type ligands (*Chem. Eur. J.* **2010**, *16*, 10844-10853). [★]



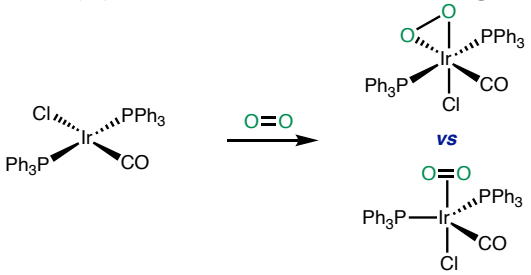
9. Marcone and Moloy (*J. Am. Chem. Soc.* **1998**, *120*, 8527-8528) studied the rate of reductive elimination from Pd(II) complexes, and observed the following rates. Hypothesize what causes the following rate trends and justify the difference between dppp and the corresponding cyclobutylated version. [★★]





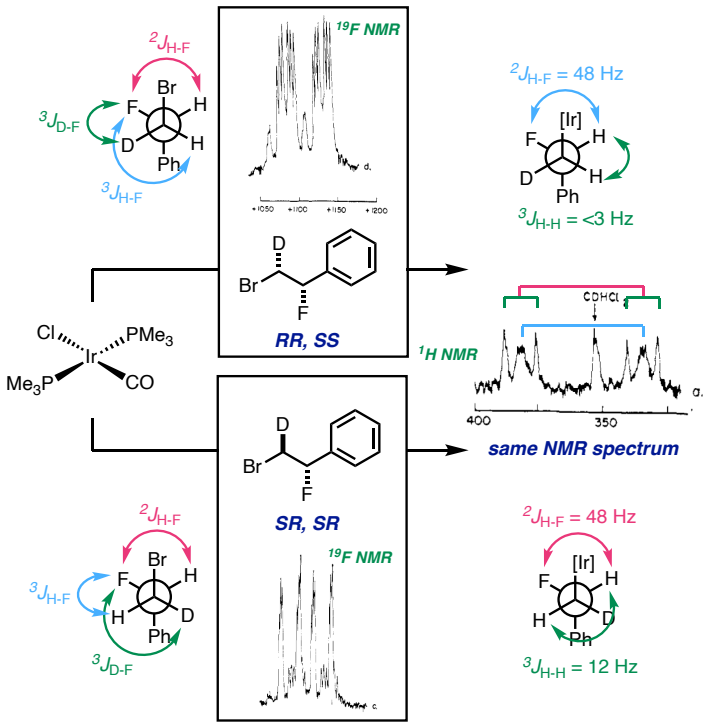
bite angle	86°	91°	102°	n.d.
k (s^{-1})	2.1 10^{-6}	5.0 10^{-5}	1.0 10^{-2}	7.4 10^{-5}

10. Suggest a method to distinguish the oxidative addition product of O_2 from the simple coordination (analogous to an olefin Dewar-Chatt-Duncanson model) (*Science* **1964**, *145*, 920-921 and *Organometallics* **2015**, *34*, 4784-4795). [★]

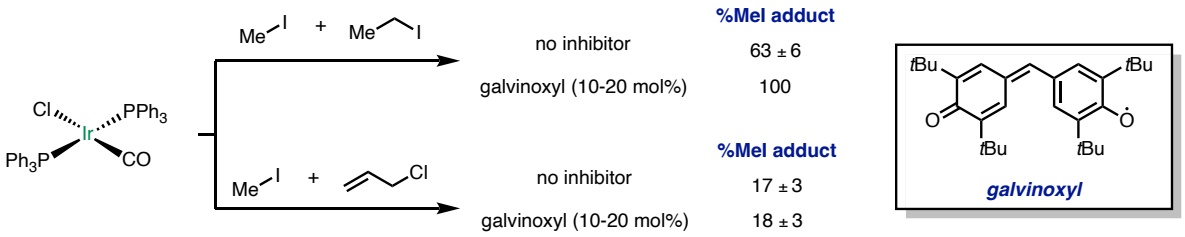


11. Osborn and co-workers (*J. Am. Chem. Soc.* **1972**, *94*, 4043-4044) studied the oxidative addition of alkyl bromide with Vaska's-type complex *trans*-[Ir(PMe$_3$)$_2$(CO)Cl]. The authors synthesized either the *RR,SS* or *RS,SR* dyad of a fluoro-

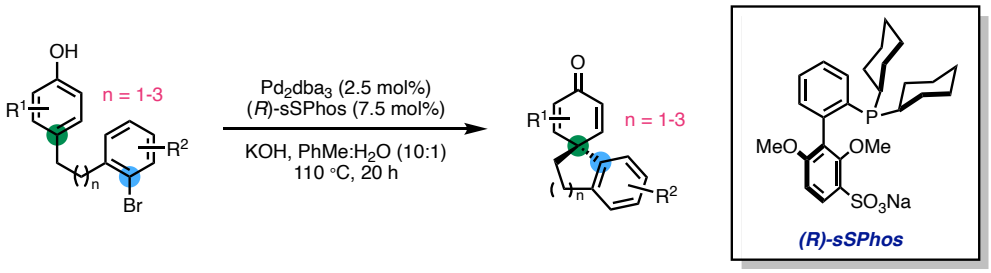
deutero-bromide, which could allow facile NMR study of the oxidative addition complex. Independent from the starting dyad, the same NMR spectrum containing resonance for both diastereoisomers was observed. Explain which oxidative addition mechanism is operative. [The different coupling constants, which allow unequivocal identification of the different species, can be estimated by using the Karplus equation] [★★★]



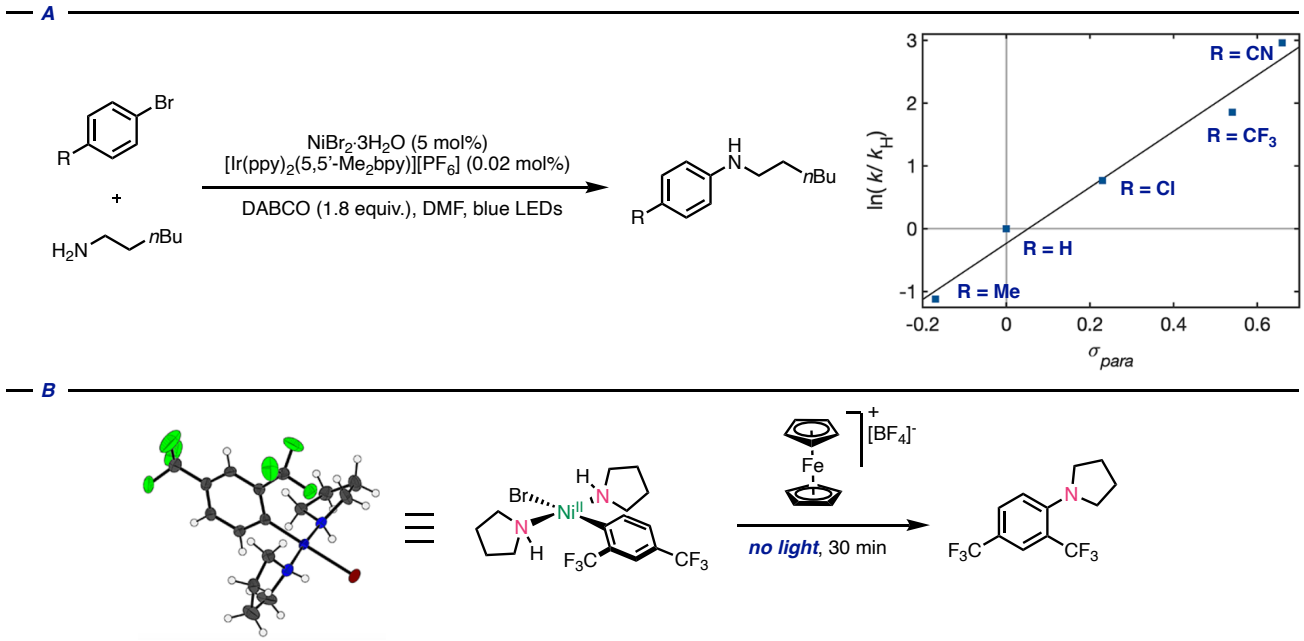
12. Labinger, Osborn and co-workers (*Inorg. Chem.* **1980**, *19*, 3236-3243) evaluated the impact of a radical inhibitor (galvinoxyl) on Vaska's-type complex *trans*-[Ir(PMe$_3$)$_2$(CO)Cl] on the competitive oxidative addition between methyl iodide and different electrophiles. Explain what the following data suggest about the mechanism of oxidative addition for ethyl iodide and allyl chloride and suggest an additional experiment to determine the exact oxidative addition mechanism for allyl chloride.. [★★]



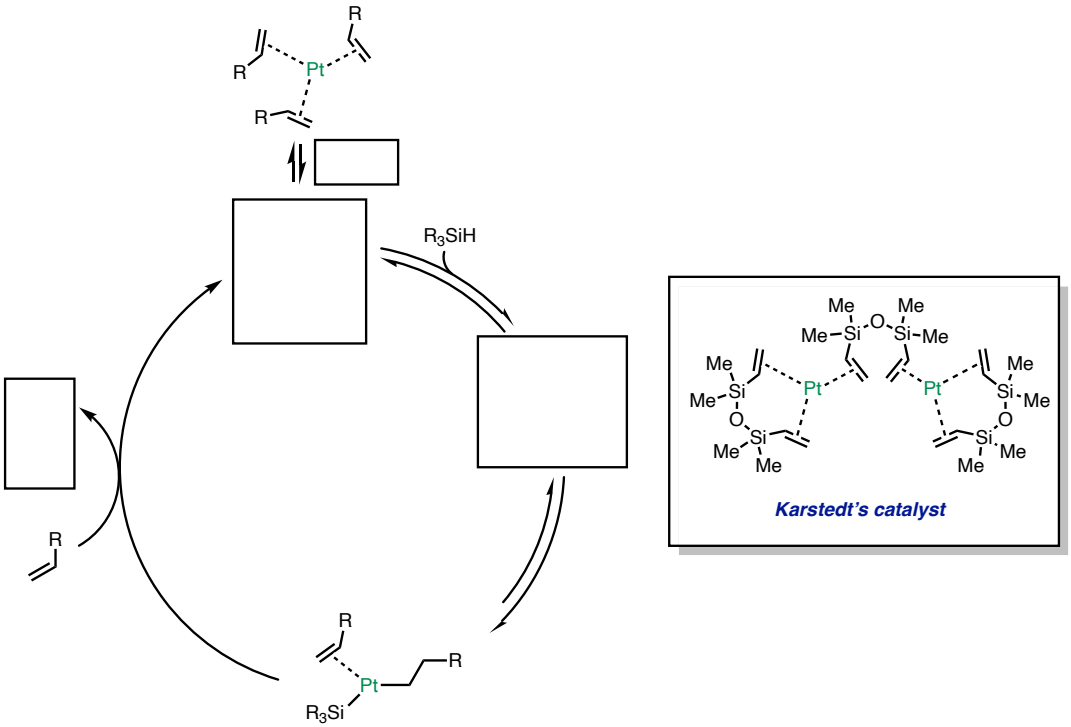
13. Phipps and co-workers (*J. Am. Chem. Soc.* **2023**, *145*, 25553-25558) reported the following enantioselective dearomative cyclization of phenols. Hypothesize the role of the ligand and draw a possible catalytic cycle. [★★]



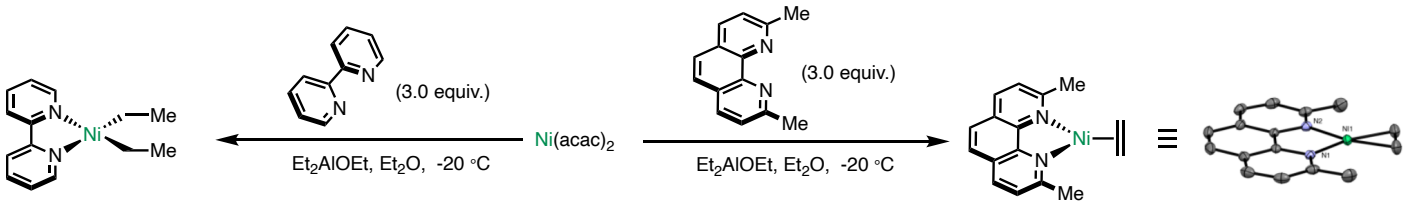
14. In 2020, MacMillan and co-workers studied a nickel metallaphotoredox catalyzed Buchwald-Hartwig-type coupling reaction (*J. Am. Chem. Soc.* **2020**, *142*, 15830-15841) and observed the following: (A) A positive *para*-Hammett correlation; (B) Upon single-electron oxidation of an intermediate Ni^{II}(aryl)(Br)(amine)$_2$ complex with ferrocenyl tetrafluoroborate (a single-electron oxidizing agent), quick product formation could be observed. Explain what these two observations suggest about oxidative addition and reductive elimination. [★★]



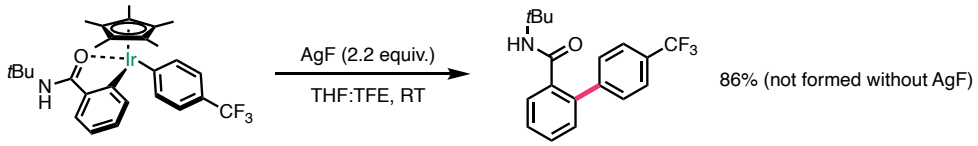
15. Olefin hydrosilylation using platinium complexes (*e.g.* Kartedt's catalyst represents an industrially important reaction to generate silicon-contaning species. Complete the mechanism below. [★] (For an example of mechanistic investigation, see *J. Am. Chem. Soc.* **1998**, *121*, 3693-3703).



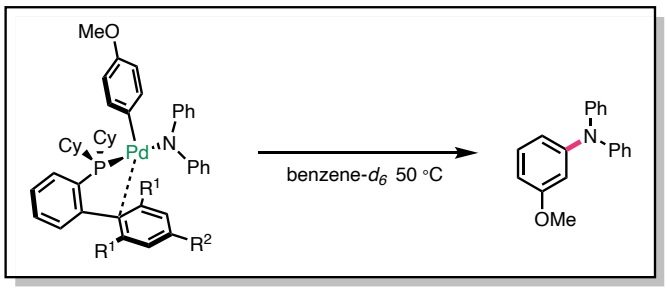
16. Martin and co-workers (*Organometallics* **2022**, *41*, 2662-2667), in the attempt to synthesize Ni^{II}-dialkyl complexes, observed the following behavior. Rationalize the effect of the two different ligands. [★★]

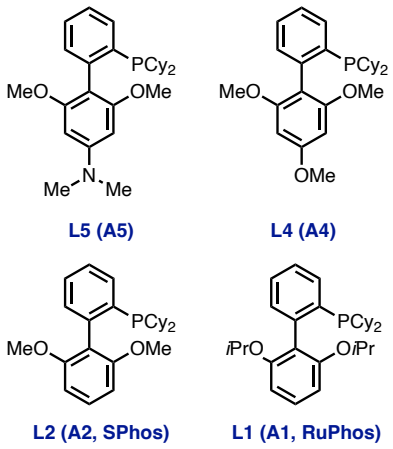


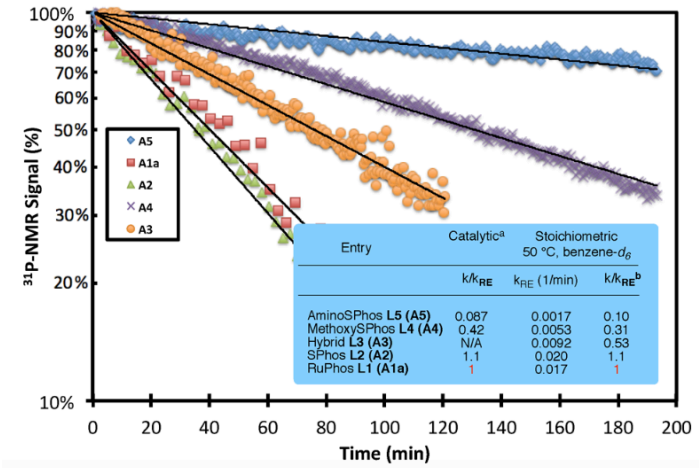
17. Chang and co-workers (*Nat. Chem.* **2018**, *10*, 218-224) reported that silver(I) oxidizing agents (*e.g.* AgF) are required to trigger reductive elimination in cyclometalated Ir(III) complexes. Rationalize the mechanism and provide an explanation to the observation. [★★]



18. Buchwald and co-workers (*J. Am. Chem. Soc.* **2016**, *138*, 12486-12493) studied the rate of reductive elimination in the eponymous coupling reaction and observed that more electron-rich ligands such as **L4-L5** caused slower rates than **L1-L2**. Provide an explanation for the observation. [★★]

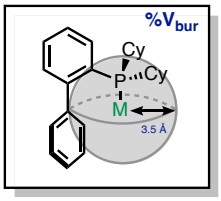




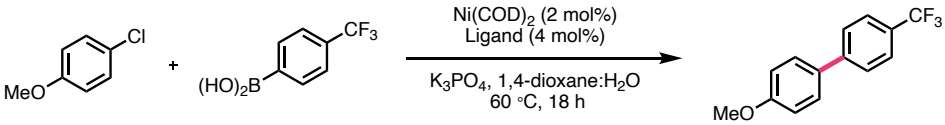


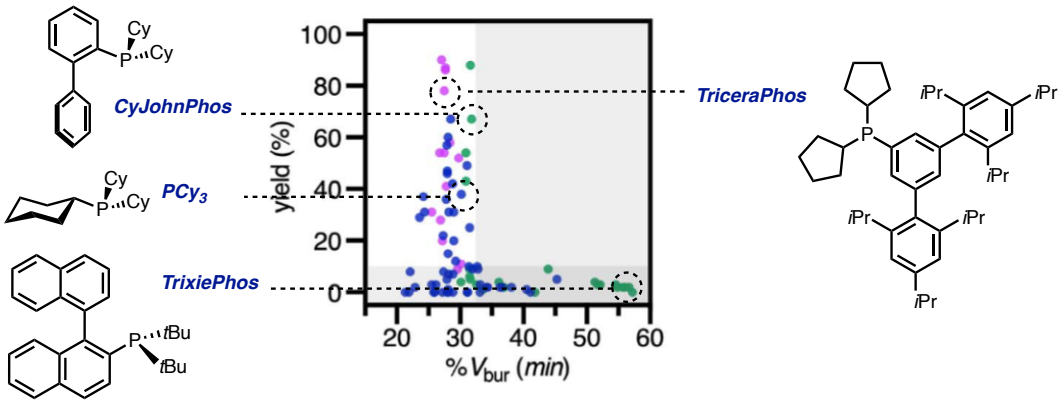
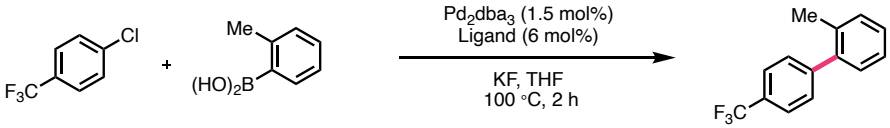
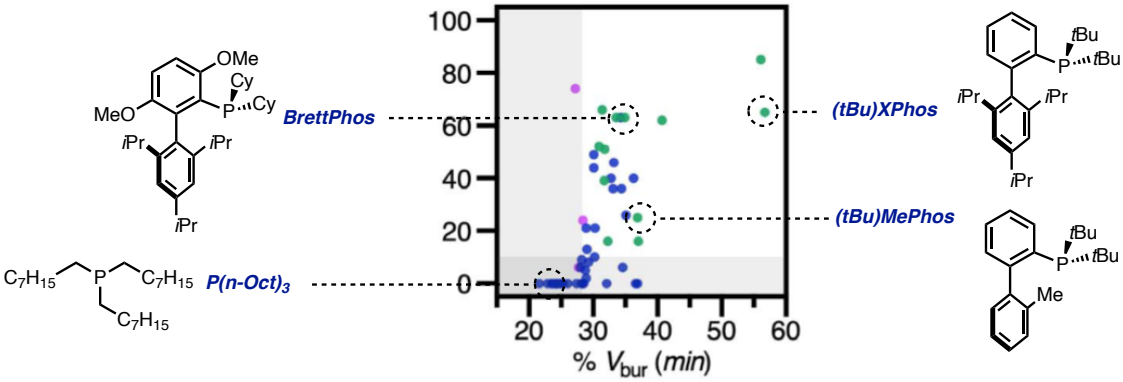
Entry	Catalytic[a] k/k_{RE}	Catalytic[a] k_{RE} (1/min)	Stoichiometric 50 °C, benzene-d_6 k/k_{RE}[b]
AminoSPhos **L5 (A5)**	0.087	0.0017	0.10
MethoxySPhos **L4 (A4)**	0.42	0.0053	0.31
Hybrid **L3 (A3)**	N/A	0.0092	0.53
SPhos **L2 (A2)**	1.1	0.020	1.1
RuPhos **L1 (A1a)**	1	0.017	1

19. Sigman, Doyle and co-workers (*Science* **2021**, *374*, 301-308) observed that nickel-catalyzed Suzuki coupling reactions with different ligands proceed when $\%V_{bur}$ (min) (the minimal energetically accessible $\%V_{bur}$, obtained from the data driven library *kraken*: *J. Am. Chem. Soc.* **2022**, *144*, 1205-1217) and the $\%V_{bur}$ (Boltz) (the average $\%V_{bur}$ obtained from the Boltzmann distribution), rather than the ligand cone angle. On the other hand, the authors observed that palladium-catalyzed Suzuki coupling reactions with different ligands proceed when $\%V_{bur}$ (min) is above a certain threshold Rationalize the experimental observations and compare the two catalytic systems. [$\%V_{bur}$ is the volume percentage occupied by the ligand within a sphere of 3.5 Å around the metal, see: *Chem. Commun.* **2010**, *46*, 841-846] [★★★]



Ni-catalyzed cross-coupling





Pd-catalyzed cross-coupling